Project Title
Water Quality Classification Using Machine Learning

Project Report submitted for the partial fulfilment of grade for the course

Machine Learning

in

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ABSTRACT

The Water Quality Classification System is a Python-based machine learning project developed to analyze and classify water quality based on key physicochemical parameters. The system uses Logistic Regression and Decision Tree classification algorithms to determine whether a given water sample is Potable or Not Potable.

The system processes important water-quality parameters, including pH, Hardness, Solids, Chloramines, Sulfate, Conductivity, Organic Carbon, Trihalomethanes, and Turbidity. Users can enter these parameters as input, after which the trained models predict the quality of the water. The project also compares the performance of both machine learning models using their classification accuracy and provides a visual representation of the Decision Tree used for classification.

Based on the experimental results, Logistic Regression achieved an accuracy of approximately 62.80%, while the Decision Tree achieved approximately 57.77%, making Logistic Regression the better-performing model among the two tested approaches. The project demonstrates the practical application of machine learning, data preprocessing, classification, model comparison, and data visualization for automated water-quality assessment.

Signature of the Guide